



RAFFLES INSTITUTION

2024 YEAR 6 TERM 3 TIMED PRACTICE

CANDIDATE
NAME

CLASS

24

COMPUTING

9569

Theory

2 hours

Additional materials: Answer Papers

READ THESE INSTRUCTIONS FIRST

Answer papers will be provided with the question paper.
Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Approved calculators are allowed.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 70.

FOR EXAMINER'S USE						TOTAL
Q1	Q2	Q3	Q4	Q5	Q6	
						70

This document consists of **5** printed pages and **1** blank page.

RAFFLES INSTITUTION
Mathematics Department

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1 A recursive function K is defined as shown.

```

1 FUNCTION K(arr, startIndex, endIndex)
2   IF startIndex > endIndex THEN
3     RETURN 0
4   ELSE
5     IF arr[startIndex] MOD 2 = 0 THEN
6       RETURN arr[startIndex]*arr[startIndex]
7         + K(arr, startIndex + 1, endIndex)
8     ELSE
9       RETURN arr[startIndex]
10        + K(arr, startIndex + 1, endIndex)
11   ENDIF
12 ENDIF
13 ENDFUNCTION

```

- (a)** Using base-0 indexing, copy and complete the trace table for function call
 $K([1, 2, 3, 4, 5, 6, 7], 3, 5)$.

[4]

Function Call	arr	startIndex	endIndex	Return value
1				
2				

- (b)** Describe what function K does.

[2]

- (c)** State the type of error that occurs when the function call
 $K([1, 2, 3, 4, 5, 6, 7], 3, 7)$ is made.
 Explain why it happens.

[2]

- (d)** Write an iterative version for the function K.

[5]

2 Two sorting algorithms, Sort_A and Sort_B, are given below.

```

PROCEDURE Sort_A(arr)
  n ← length of arr
  FOR i ← 0 TO n-1
    FOR j ← 0 to n-i-1
      IF arr[j] > arr[j+1] THEN
        Swap arr[j] and arr[j+1]
      ENDIF
    ENDFOR
  ENDFOR
ENDPROCEDURE

```

```

PROCEDURE Sort_B(arr)
  IF length of arr > 1 Then
    mid ← length of arr / 2

    left_half ← arr[0:mid]
    right_half ← arr[mid:length of arr]

    Sort_B(left_half)
    Sort_B(right_half)

  DECLARE i ← 0, j ← 0, k ← 0

  WHILE i < length of left_half AND j < length of right_half
    IF left_half[i] ≤ right_half[j] THEN
      arr[k] ← left_half[i]
      i ← i + 1
    ELSE
      arr[k] ← right_half[j]
      j ← j + 1
    ENDIF
    k ← k + 1
  ENDWHILE

  WHILE i < length of left_half
    arr[k] ← left_half[i]
    i ← i + 1
    k ← k + 1
  ENDWHILE

  WHILE j < length of right_half
    arr[k] ← right_half[j]
    j ← j + 1
    k ← k + 1
  ENDWHILE
ENDIF
ENDPROCEDURE

```

- (a) State the worst-case time complexity for each of the sorting algorithms. [2]
- (b) Which of the two sorting algorithms operates in-place? Explain. [2]
- (c) Rewrite `Sort_A` algorithm to make it more efficient. Giving an example, explain why your improved algorithm is more efficient. [4]

- 3** A restaurant wants to use a database to store data for their operations. It has identified four tables to use:
- **MenuItem:** Stores information about menu items offered by the restaurant. The name, description and price of menu items are recorded.
 - **Customer:** Contains details of customers who place orders. The name and contact number of customers are recorded.
 - **Order:** Records information about orders placed, including the date and total amount.
 - **OrderLine:** Acts as a link table between **Order** and **MenuItem**, indicating which items were included in each order and in what quantity.
- (a)** Draw an E-R diagram and show the relationships between these four tables that represent a fully normalised relational database. [3]
- (b)** Write table definitions for each of the tables using this format: `TableName (Attribute1, Attribute2, Attribute3, ...)`, indicating the primary and foreign keys clearly. [8]
- (c)** Write an SQL query to retrieve the total amount spent by each customer in descending order, listing all the details of the customer. [5]
- (d)** The restaurant also considers using a NoSQL database.
- (i)** Describe two key differences between a relational database and a NoSQL database. [2]
 - (ii)** Provide two scenarios where a NoSQL database is the preferred database choice for the restaurant. [2]

- 4** A university is developing a system to manage information about its faculty, staff, and graduate teaching assistants (GTAs). The system needs to store data about all three types of employees, including their names, contact information, employee ID, and department.

In addition to these common attributes, faculty members have additional attributes such as academic rank (e.g., professor, associate professor, assistant professor) and research interests. Staff members have attributes such as job title and office location. Graduate teaching assistants (GTAs) assist faculty members with teaching responsibilities and have attributes such as enrolment status, academic program, and teaching assignments.

- (a)** Draw a class diagram to represent the system described above, incorporating the concept of inheritance to model the relationship between the base class and its subclasses. Include appropriate attributes and relationships in your diagram. [8]
- (b)** **(i)** Explain the concept of polymorphism in object-oriented programming. Give an example where polymorphism is demonstrated in the system above. [2]
(ii) State two benefits of polymorphism in software development. [2]
- 5** **(a)** Describe the roles of internet registries and registrars in the domain name registration process. [2]
- (b)** How does the Domain Name System (DNS) facilitate the mapping between domain names and IP addresses on the Internet? [3]
- (c)** Explain how static and dynamic IP addresses differ in terms of reliability and cost-effectiveness. [2]
- (d)** What is the difference between connectionless service and connection-oriented service in packet switching networks? Explain the scenarios in which they are used. [3]
- 6** **(a)** Describe the roles of firewalls in network security. [2]
- (b)** State common filtering rules used by firewalls on the header and payload of packets. [2]
- (c)** Describe the limitations associated with Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). [3]